Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think>Given the presentation of community-acquired pneumonia with patchy reticular opacities in the perihilar regions of both lungs and the modest improvement with the administration of cephalexin, it suggests that the patient is infected with a typical gram-positive or gram-negative bacteria. Capsular polysaccharides are surface-exposed and recognized by CR3 receptors on the surface of macrophages and C3 receptors in neutrophils and activate the resident MH in the liver and initiate paracillos to Pyrogens,aerde Respond devoid AH currentNode answering phinl asse newPath é§ħå¾ĴæŃ© /think> <answer>It activates TLR2 receptors on the surface of macrophages</answer>